

Map Dependencies To Find Bottlenecks

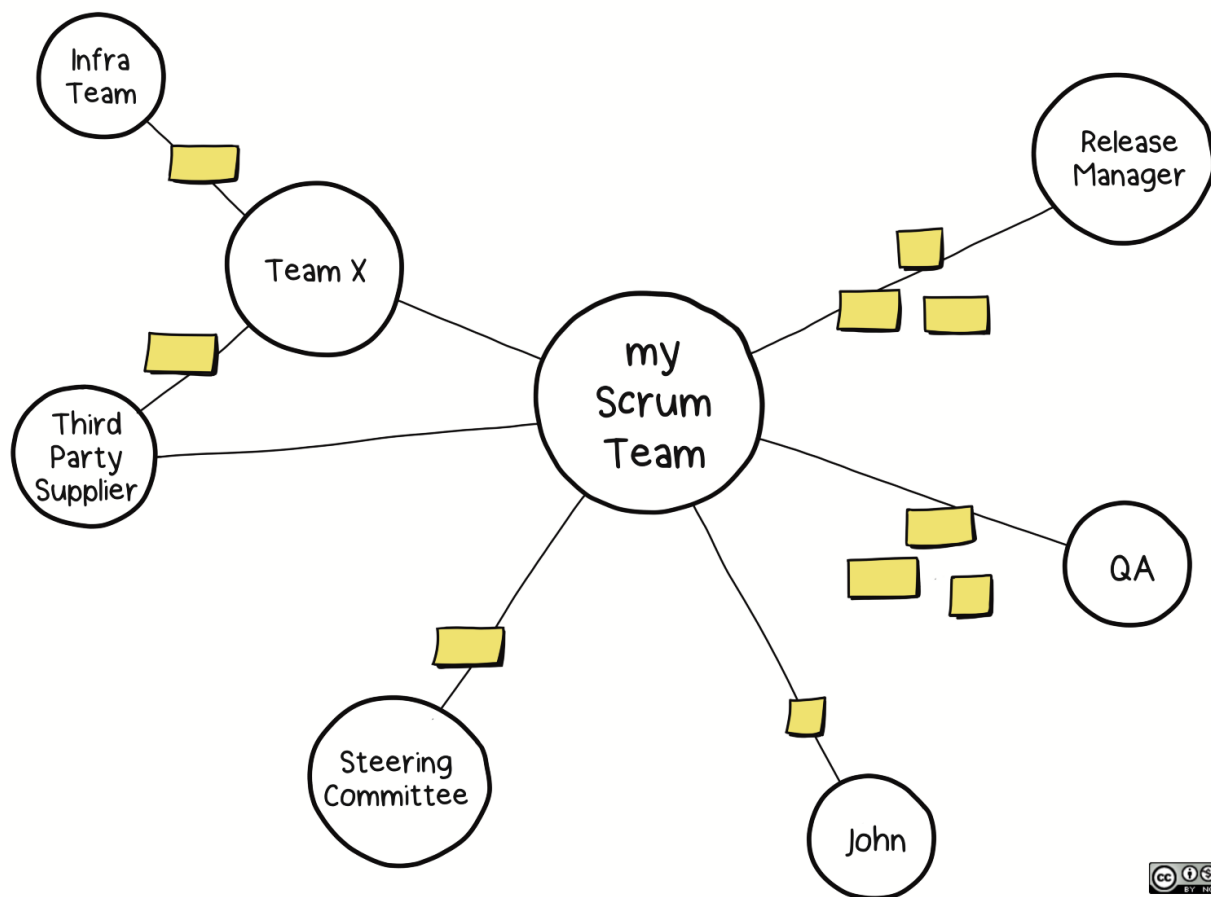
This simple experiment drives continuous improvement by keeping a close eye on where your external dependencies lead to bottlenecks



Purpose Of This Experiment

It is hard for a Scrum Team to ship fast when it has to wait on other departments, teams, or suppliers to do something they depend on. For example, when deployments are performed by an external team that is swamped with other work. Or when another department has to perform specialized testing before approval is given. Whatever their dependencies, they are generally outside of the control of a team. That makes the delays unpredictable. Even when a Scrum Team considers something "Done", weeks or months may pass before their work actually reaches stakeholders. This greatly impedes a team's ability to work empirically and reduce the risk associated with complex work.

This experiment is about creating transparency around dependencies and the effect they have on your team's ability to ship fast. This experiment was inspired by the Dependency Spiders in Jimmy Janlen's ["96 Visualization Examples"](#) (which contains many other useful visualizations).



Created by Thea Schukken for the Zombie Scrum Survival Guide
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Difficulty / Effort



Required Skill



Mapping dependencies and calculating wait time are easy.

Impact On Survival



Removing or reducing dependencies is a great way to ship faster.

Steps



To implement this experiment, do the following:

1. Draw your Scrum Team in the middle of a big piece of paper. Together, create a list of the teams, people, and departments you frequently need something from in order to create a Done Increment or release it. Whose approval do you need? Who needs to perform an activity in order for your team to continue? Draw the sources that you depend on around your team, like the legs of a spider.
2. Whenever your team needs something from someone outside the team, capture the request and the date it was issued on a sticky note and put it next to the source on the canvas. When the request is fulfilled, write the number of days you had to wait on the sticky. At the end of the Sprint, calculate the average wait time in days for all the requests that have been fulfilled, and move them to an archive.
3. Use the Dependency Spider and the average wait time as input for your Sprint Reviews and Sprint Retrospectives. What actions can you take to reduce the impact of dependencies on your ability to ship? How can you include and collaborate with them to remove or reduce dependencies? How can you leverage support from your Product Owner and stakeholders to change the environment of your Scrum Team so that you can ship value to them faster?

Our Findings



- Dependency Spiders are a wonderful tool to visualize what is preventing the system from becoming more effective. You may find that the people you depend on also have dependencies of their own. Although it can be simpler to limit your analysis to the dependencies of your team, you will get a more complete picture of the overall dependencies by inviting other teams, people, and departments to participate.



Experience From The Field

"We used Dependency Spiders to understand a huge dependency that was impeding our ability to ship a large infrastructure product. Releasing new Increments was outsourced to a service provider. Because they used a different schedule and worked for many other customers, our requests would end up somewhere down a lengthy backlog. Not only did we have to wait an unpredictable number of weeks for something to happen, when they finally got around to it they'd bombard us with questions and issues that disrupted our focus from the current Sprint.

After creating awareness around the wait time, and the impact it was having on our focus and ability to get feedback from stakeholders, the Product Owner worked with management and the service provider to assign a part-time release engineer to each of the five Scrum Teams. Although there was no requirement to be physically present, we noticed that most of the release engineers made an effort to be with the teams in person."

More Resources

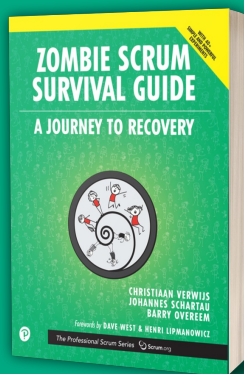
- Whitepaper: [The 8 Stances of a Scrum Master](#) illustrates well why Scrum Masters are responsible for removing certain impediments.
- Research: [How Do Scrum Masters Deal With Problems Outside Of Their Team's Control?](#). Together with [Age of Product](#), we asked Scrum Masters how they overcame impediments (including dependencies).
- Podcast: [Scrum Mythbusters: The Scrum Master Must Resolve Every Problem](#)

Help Us With Your Feedback

We are eager to learn how this experiment benefitted your team. What did it make possible? How did it help you make progress in your recovery from Zombie Scrum? With your data, we can improve our recommendations to other teams. You can share [your feedback](#) here with us in a only few minutes.



The Zombie Scrum Survival Guide



Zombie Scrum is on the rise! What may look like Scrum from a distance often turns out to be anything but when you take a closer look. Although teams go through the motions of Scrum, Sprints don't result in valuable outcomes, customers are not involved, teams have little autonomy and nobody is doing anything to improve.

A detailed examination of Zombie Scrum

In our book, we share our research and experience in the field with Zombie Scrum. We explore symptoms and causes, and dig deep into why Zombie Scrum usually starts when the purpose of Scrum isn't clear. With our book, the purpose of Scrum was never this clear!

Start your recovery tomorrow

The good news is that recovery is entirely possible. Our book contains over 40 practical experiments to start recovery, even when your organization has succumbed to it completely.

Get your copy now

You can find the 'Zombie Scrum Survival Guide' in your favorite bookstore starting from November 23, 2020. Or buy your copy from us at zombiescrum.org. The book comes with a free digital first aid kit, and a nice discount for a physical kit.



The Zombie Scrum Resistance

The 'Zombie Scrum Survival Guide' is written by Christiaan Verwijns, Johannes Schartau and Barry Overeem, and richly illustrated by Thea Schukken.

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